

Principles for Archiving Metadata

1. Maintain and do not alter the semantic meaning of the questionnaire

Principle 1 is the overall aim of CLOSER's **metadata ingest project** and therefore the most important. CLOSER's target is to generate a search engine containing every instrument used by **9 individual longitudinal cohort studies**. The search engine, often referred to as the **Metadata Discovery Platform (MDP)**, is designed for examining the logical flow and semantic meaning of the instruments. Therefore *Principle 1* has to be **kept to at all times**. CLOSER also intends that the metadata

documented is capable of being shared with other DDI compliant organisations, hence *Principle 1* also ensures that CLOSER produces **consistent and comparable** metadata.

The practice of keeping to *principle* requires decisions to be made as to what questionnaire elements provide meaning. For example it was decided that **bold font provided no semantic meaning** and therefore CLOSER is not documenting the weight of the font. The order of multiple

choice options was deemed **relevant** and therefore special care is taken to **preserve the order** within the documentation.

The three following principles were conceived purposefully to give **structure and guidance** as to how the first principle should be **followed at all times and in all situations**.

8 Does the mother care for children at home (including adopted and foster children) ?

Yes1
No2
Not known0

2. Do not correct the questionnaire

Principle 2 is the second most significant principle for ingesting metadata. It is fairly common to find what seem like mistakes in the questionnaire design; these can range from **typos** (e.g. 'marital status' instead of 'marital status') to **impossible condition logic** (e.g. asking when someone gave up smoking, within a branch of non-smokers). Any mistakes within the questionnaire could have **altered the data being collected**, and therefore it is important to avoid correcting the metadata while

documenting. Also, what seems like a mistake always has the potential of being done purposefully and therefore potential **mistakes should be verified**.

In the case of a typo, the misspelt word can be aliased within the MDP to allow effective searching (e.g. searches looking for the 'marital', would also find aliased questions with the word 'marital').

There are rare situations where *Principle 2* must be violated in order to follow *Principle 1* and document meaningful metadata. A

real-world example of this is when a code value is accidentally **printed with the wrong value**. There is no easy way to alias code values and CLOSER's documentation would suggest that there are two distinctly different codes, while the dataset would only refer to one of the codes. In this case, documenting the mistake **would mislead the user** and violate *Principle 1*.

102. Have you ever written that indicate that something happened in the past; please mark in each column which he never and/or understands

	(1) Yes, he says	(2) Yes, he understands
(1) wife	☐	☐
(2) blow	☐	☐
(3) held	☐	☐
(4) coat	☐	☐
(5) made	☐	☐
(6) ran	☐	☐
(7) sat	☐	☐
(8) saw	☐	☐
(9) took	☐	☐
(10) went	☐	☐

3. Only record what is contained within the questionnaire

Principle 3 is the third most significant principle and therefore should be broken only when doing so maintains *Principles 1 & 2*.

It is important while recording historic metadata to **refrain from adding additional information** that is not within the questionnaire. A seemingly simple concept, but there are situations, particularly in older questionnaires, where the questionnaire does not provide all of the information to document meaningfully or

to **generate valid DDI**. A real-world example of when not to add to the metadata is when a questionnaire asks

"how many?". 3.1

Often this question and similar questions can be found within a condition, immediately following a question similar to

"Do you own a car?". 3.2

Question 3.1 is **largely meaningless** without being able to see question 3.2, which makes it tempting to copy question 3.2's text and concatenate it with question 3.1

creating

"Do you own a car? how many?". 3.3

Doing this **violates Principle 1**, and is therefore forbidden. The solution to maintain context for question 3.1 can be found by **accurate routing using conditional constructs**. For example question 3.1's universe would contain "Answered yes to 'Do you own a car?'".

An example where the metadata must be added in order to maintain *Principle 1* is when a questionnaire uses an **arrow to**

denote a condition. It is impossible to document an arrow literally and leaving it out of the documentation altogether **changes the logical flow** of the questionnaire. Therefore, text representing the arrow's meaning has to be added.

11 Does this child have Special Educational Needs (SEN)?
Tick ONE box ONLY

Yes ☐ → Go to 12
No ☐ → Go to 14

4. Do not allow the data recorded to inform the metadata archiving

Principle 4 is the final and least significant principle. Whilst documenting the structure, flow and intent of a questionnaire, it may seem harmless to **consult the collected data** in order to better understand the questionnaire being recorded. This practice, however, should also be **avoided**. The aim of the ingest programme is to record the instruments used for data collection as accurately as possible. Using information that was **created after the collection** event alters the perception and understanding of the instrument.

A very basic real-world example of this can be found when a question offers a set of multiple choice options and an "other (specify)" option. If the same response is often specified within the other option, then it is relatively common to code that answer in addition to the originally offered options. It is important not to document this additional coded answer, because it was not offered to the respondents and therefore potentially had an **effect on the collected data**.

The most common situation where breaking *Principle 4* is valid is when *Principles 2 or 3* must be broken. For example, in the situation where it is most appropriate to correct the questionnaire, it is obviously **vital to check** whether the mistake was intentional or not and if the mistake had a **distinct effect on the collected data**.

Who undertook the antenatal care?

Hospital medical staff. AC193A...1
L.H.A. M.O. AC193B...2
G.P. AC193C...3
Midwife Hospital AC193D...4
G.P. Unit AC193E...5
Domiciliary AC193F...6
Other (specify) AC193H...7
CONSULTANT AC193G...